

Brian W. Smith

PRINCIPAL SITE RELIABILITY ENGINEER · DISTRIBUTED SYSTEMS RELIABILITY · OBSERVABILITY AT SCALE

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25+ YEARS IN PRODUCTION	100+ KUBERNETES CLUSTERS OPERATED AT SCALE	SLO SLI / ERROR BUDGET FRAMEWORKS	7,000+ MICROSERVICES OPERATED AT SCALE
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SUMMARY

Principal SRE at Red Hat with 25+ years of enterprise infrastructure experience, specializing in large-scale reliability engineering, distributed systems observability, and operational excellence. Deep expertise in SLI/SLO frameworks, incident management, and cloud-native platform engineering across OpenShift/Kubernetes, AWS, and Terraform IaC. Proven track record of building and operating platforms at scale — from greenfield infrastructure to 100+ cluster multi-region environments — while driving reliability improvements and automation across cross-functional engineering teams. Currently leading platform reliability, secret management engineering, and AI-powered operational automation across the ROSA (Red Hat OpenShift Service on AWS) Regional Platform.

TECHNICAL SKILLS

▸ OpenShift / Kubernetes (CKA)	▸ Splunk / Dynatrace / Prometheus / Grafana	▸ Incident Response / SLI/SLO / Error Budgets
▸ AWS Multi-Region / PrivateLink	▸ Terraform / Infrastructure as Code	▸ Bash / Python / Go
▸ GitOps / ArgoCD / Helm / Kustomize	▸ CI/CD (Tekton, Jenkins, GitLab)	▸ HashiCorp Vault / Secrets Management
▸ Zero Trust / FIPS 140-2	▸ Keycloak / IAM / SSO	▸ Claude AI / MCP / Agent SDK
▸ FedRAMP High ATO	▸ NIST 800-53 Rev 5	

EXPERIENCE

Principal SRE (January 2026 – Present)

- Implemented 13+ NIST 800-53 Rev 5 control families (AU, AC, SC, SI, RA, IA, CA, CP) for FedRAMP Moderate compliance across the ROSA Regional Platform, mapping each against HCP (Hosted Control Plane) Terraform modules to produce a gap-closed control inventory.
- Deployed a Thanos/Prometheus/Alertmanager/Grafana observability suite via parameterized IaC, enabling single-variable regional rollouts across all ROSA environments.
- Designed and deployed a Secret Management framework via AWS Secrets Manager and Vault sidecar injection, enforcing least-privilege IAM policies per service and eliminating plaintext secrets across all repositories.
- Built a Python/DynamoDB Sandbox account leasing system provisioning scoped IAM roles and SCPs with auto-reclaim, cutting environment setup from hours to minutes.
- Architected a 3-tier AWS CodePipeline GitOps hierarchy with a meta-pipeline provisioner dynamically managing per-cluster pipelines from Git commits, paired with ArgoCD ApplicationSet Matrix Generators for zero-config multi-cluster delivery.
- Eliminated VPC peering and Transit Gateway coupling by engineering zero-trust cross-cluster messaging over AWS IoT Core MQTT; deployed ECS Fargate bootstrap-then-handoff for fully private EKS initialization and a 17-module reusable Terraform library; built ephemeral CI environments (provision → e2e → teardown via Prow).
- Automated FedRAMP compliance evaluation via a Claude Sonnet sub-agent pipeline assessing ~700 Moderate and ~100 High-only NIST 800-53 Rev 5 controls — 7-step workflow (clone → classify → analyze → report → PDF → Drive → Jira/PR), 12 artifacts per run, with Trivy CVE integration auto-downgrading Met controls on CRITICAL/HIGH findings.
- Led cross-vendor FIPS validation for OCP on EKS, using negative-control testing on Go binary dlopen dispatch paths to `/proc/sys/crypto/fips_enabled` to distinguish infrastructure-level from binary-level enforcement; collaborated with Red Hat cryptographic maintainers on KAT tests for NIST SP 800-131A AES/RSA/ECDSA compliance.

Principal SRE - Team Lead (January 2025 – January 2026)

- Led FedRAMP CONMON remediation across a fleet of clusters; resolved SRE emergency access blockers by integrating backplane-cli/backplane-api with FedRAMP-compliant auth protocols.
- Architected AWS PrivateLink for ROSA HCP across all GovCloud regions with cross-account VPC Peering.
- Led the migration of 8 ROSA clusters across 2 GovCloud regions from classic to Hosted Control Plane (HCP) architecture, driving cross-functional coordination across multiple Red Hat engineering and product teams to deliver a successful FedRAMP compliance assessment.
- Developed and delivered 10+ technical training sessions on ROSA HCP to SRE teams across all GovCloud regions, driving large-scale adoption of a newly launched managed control-plane architecture and reducing time-to-productivity for engineers onboarding to the platform.
- Deployed Dynatrace Managed Service across FedRAMP environments with OneAgent log forwarding; engineered a Keycloak → Splunk log pipeline for centralized audit trails and real-time authentication security monitoring; authored SCR architecture diagrams and ADRs for External DNS Operator and HCP authorization.

Senior SRE Functional Lead (December 2023 – January 2025) · Senior SRE (November 2021 – December 2023)

- Led a team of 9 SREs; defined and drove agile epics aligned with business priorities — CVE mitigation, annual FedRAMP assessments, Significant Change Requests, and operational toil reduction.
- Deployed Red Hat build of Keycloak (RHBK) to production, authored six FedRAMP Boundary Architecture Diagrams, and engineered real-time synthetic availability monitoring for the identity platform.
- Automated AWS Config, GuardDuty, and SecurityHub via Terraform to establish the FedRAMP security baseline; authored the SSP Access Control validation epic for the ROSA Regional Platform.
- Led GoAlert deployment as the FedRAMP on-call platform from PoC through VPCE network architecture to production, replacing PagerDuty and reducing on-call escalation latency.

- Built the greenfield FedRAMP High GovCloud platform on AWS; established Terraform as the IaC provisioning foundation; led ATO workstreams including annual FedRAMP assessment support.
- Architected Transit Gateway hub-and-spoke topology for GovCloud; drove global VPC Flow Logs rollout and WAF compliance audit across all high-compliance regions.

OneStream Software

February 2021 – November 2021

FIRST DEVOPS ENGINEER

- As the organization's first DevOps engineer, migrated 70+ repositories from TFS to Azure DevOps and established CI/CD foundations including SonarQube SAST, GitFlow branching, and Azure Pipelines.

Immuta

August 2020 – February 2021

CLOUD ENGINEER

- Deployed and configured Immuta's data security platform for enterprise customers across Databricks, Athena, Snowflake, and Redshift, applying Kubernetes expertise to accelerate onboarding across data warehouse environments.

Fiserv

March 2017 – August 2020

SENIOR SYSTEM ENGINEER LEAD – CLOUD PLATFORM

- Designed and built 16 Pivotal Cloud Foundry foundations supporting 7,000+ microservices; deployed Kubernetes into Azure using Pivotal Kubernetes Service (PKS/TKGI).
- Led Fiserv's PaaS deployment to Azure and authored the enterprise Terraform infrastructure; presented the hybrid-cloud Terraform architecture at SpringOne 2019.
- Built a self-service Kubernetes cluster provisioning platform (web form → Git → pipeline → Helm-deployed Prometheus/Grafana stack + expiration cleanup); designed centralized SLI dashboards across 14 PAS foundations using Grafana, AlertManager, and Prometheus.
- Formed and trained the Cloud Platform team, establishing agile delivery practices across the enterprise Kubernetes and PaaS program.

Earlier Roles

1998 – 2008

GRANGE INSURANCE (SENIOR SYSTEM ENGINEER) • NETJETS (SR. NETWORK ADMIN) • BMW FINANCIAL (CRM ADMIN) • CARDINAL HEALTH (SR. MESSAGING ENGINEER) • WD PARTNERS (IT ADMIN)

- 25+ years of progressive infrastructure experience spanning messaging, networking, virtualisation, enterprise AD/Exchange, and automation — from first ISP support role (BellSouth.net) through senior engineering leadership.

CERTIFICATIONS

Certified Kubernetes Administrator (CKA)	CKA-2000-008308-0100
Azure DevOps Engineer Expert (AZ-400)	December 2020
Azure Solutions Architect Expert (AZ-303/304)	October 2020
Azure Administrator Associate (AZ-104)	October 2020

EDUCATION & NOTABLE

Franklin University — Computer Science	Columbus, OH
Eagle Scout — Order of the Arrow	Boy Scouts of America
SpringOne 2019 Speaker	Pivotal Platform / Azure
Open Source Contributor	github.com/theautoroboto